

phone. After the user finishes, the Speech Recognition API will fire the 'onresult' event and make the results of the recognised speech available as a JavaScript object.

Set a language

You can set a language or dialect from the vast number of available languages.

```
var recognition = new webkitSpeechRecognition();
recognition.lang = "en-GB";
```

What about OTHER browsers?

Support for other browsers is almost non-existent for speech recognition. However, with HTML5 we can expect more and more advances in the field of speech recognition.

The HTML5 Speech Recognition API, defined as an experimental one by Mozilla's MDN, allows JavaScript to have access to a browser's audio stream and convert it to text. Needless to say that microphone access is mandatory. If your site is on 'HTTPS' then your browser remembers these permission settings explicitly.

The other JS libraries, like PocketSphinx (<http://dgit.in/PcktSphnx>) works on Firefox, Edge as well as Chrome. However, the accuracy of these are not as high as Annyang. But the very existence of these gives us hope that speech recognition will soon become an integral part of web applications and will be supported extensively by other browsers.

Security

Pages served on HTTP requires permission each time they want to make an audio capture in a similar way to requesting access to other items via the browser. Pages on HTTPS do not have to repeatedly request access and the browser remembers the access provided to it.

The Chrome API interacts with Google's Speech Recognition API so all of the data is going via Google and whoever else might be listening. Also once the permission is granted, the application can keep listening until the browser window is closed or until it is explicitly stopped.

In the context of JavaScript, the situation is slightly different as the entire page has access to the output of the audio stream.

Hence when it comes to JavaScript you obviously have to handle the audio differently.

OTHER USES OF CHROME's Web Speech API

Web Apps:

- **Speechnotes** – Speechnotes is a dictation platform that has its own website and an Android app. The platform comes with handy assists regarding insertion of punctuation and capitalisation. And it works perfectly well offline. <http://dgit.in/SpchNotes>
- **Dictation** – This one is also a dictation app that comes with its own Chrome app. This was developed by Amit Agarwal. The platform uses the x-webkit-speech attribute of HTML5 that is only implemented in Google Chrome <http://dgit.in/Dictn>
- **Chrome Demo** – This one's the standard demo of the Web Speech API on Google Chrome. <http://dgit.in/ChrmDemo>

SOME OTHER VOICE RECOGNITION JAVASCRIPT LIBRARIES

- **Artyom** – Artyom.js is an useful wrapper of the speechSynthesis and webkitSpeechRecognition APIs. Besides, artyom.js also lets you to add voice commands to your website easily too! <http://dgit.in/ArtyomJS>
- **PocketSphinx** – Pocketsphinx.js is another speech recognition library written entirely in JavaScript and running entirely in the web browser. It does not require Flash or any browser plug-in and does not do any server-side processing. It makes use of Emscripten to convert PocketSphinx from C, into JavaScript. Audio is recorded with the getUserMedia JavaScript API and processed through the Web Audio API. <http://dgit.in/PcktSphnx>

Credits

- Tal Ater – the author of the annyang library <http://dgit.in/TalAter>
- Mac Terminal CSS inspiration from codepen by <http://dgit.in/DenCPen>
- For the basic javascript of ELIZA - George Dunlop and <http://dgit.in/PeccviCm> >

*Nodes of Interest



*HTML5 API tutorial

>>A short but handy guide into the Speech Recognition API from HTML5.

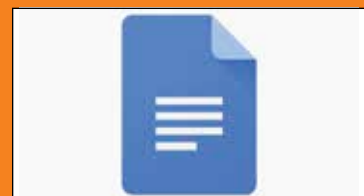
<http://dgit.in/ShpShd>



*Mozilla Development

>>Mozilla's documentation on the Mozilla Developer network platform to for the Web Speech API.

<http://dgit.in/MzlaSpch>



*Google Docs Add-on

>>If you are a Google Docs loyalist, this add-on, an implementation of the Web Speech API, is a blessing.

<http://dgit.in/GDocsAddon>